

- Tablet 5 mg scored, 10 mg scored, 15 mg multiscored, 30 mg multiscored

How to Dose

- Initial 15 mg twice a day; increase in 5 mg/day increments every 2–3 days until desired efficacy is reached; maximum dose generally 60 mg/day



Dosing Tips

- Requires dosing 2–3 times a day for full effect
- Absorption is affected by food, so administration with or without food should be consistent

Overdose

- No deaths reported in monotherapy; sedation, dizziness, small pupils, nausea, vomiting

Long-Term Use

- Limited data suggest that it is safe

Habit Forming

- No

How to Stop

- Taper generally not necessary

Pharmacokinetics

- Metabolized primarily by CYP450 3A4
- Elimination half-life approximately 2–3 hours
- Absorption is affected by food



Drug Interactions

- Use with caution with MAOIs, including 14 days after MAOIs are stopped (for the expert)
- CYP450 3A4 inhibitors (e.g., fluoxetine, fluvoxamine, nefazodone) may reduce clearance of buspirone and raise its plasma levels, so the dose of buspirone may need to be lowered when given concomitantly with these agents
- CYP450 3A4 inducers (e.g., carbamazepine) may increase clearance of buspirone, so the dose of buspirone may need to be raised
- Buspirone may increase plasma concentrations of haloperidol
- Buspirone may raise levels of nordiazepam, the active metabolite of diazepam, which may result in increased symptoms of dizziness, headache, or nausea



Other Warnings/Precautions

- None

Do Not Use

- If patient is taking an MAOI (except as noted under Drug Interactions)
- If there is a proven allergy to buspirone

Special Populations

Renal Impairment

- Use with caution
- Not recommended for patients with severe renal impairment

Hepatic Impairment

- Use with caution
- Not recommended for patients with severe hepatic impairment

Cardiac Impairment

- Buspirone has been used to treat hostility in patients with cardiac impairment

Elderly

- Some patients may tolerate lower doses better



Children and Adolescents

- Studies in children age 6–17 do not show significant reduction in anxiety symptoms in generalized anxiety disorder (GAD)
- Safety profile in children encourages use

Pregnancy

- Effective June 30, 2015, the US FDA requires changes to the content and format of pregnancy and lactation information in prescription drug labels, including the elimination of the pregnancy letter categories; the Pregnancy and Lactation Labeling Rule (PLLR or final rule) applies only to prescription drugs and will be phased in gradually for drugs approved on or after June 30, 2001
- Controlled studies have not been conducted in pregnant women
- Animal studies have not shown adverse effects
- Not generally recommended in pregnancy, but may be safer than some other options

Breast Feeding

- Some drug is found in mother's breast milk
- Trace amounts may be present in nursing children whose mothers are on buspirone
- If child becomes irritable or sedated, breast feeding or drug may need to be discontinued

The Art of Psychopharmacology

Potential Advantages

- Safety profile
- Lack of dependence, withdrawal
- Lack of sexual dysfunction or weight gain

Potential Disadvantages

- Takes 4 weeks for results, whereas benzodiazepines have immediate effects

Primary Target Symptoms

- Anxiety



Pearls

- * Buspirone does not appear to cause dependence and shows virtually no withdrawal symptoms
- May have less severe side effects than benzodiazepines
- * Generally lack of sexual dysfunction
- Buspirone may reduce sexual dysfunction associated with GAD and with serotonergic antidepressants
- Sedative effects may be more likely at doses above 20 mg/day

- May have less anxiolytic efficacy than benzodiazepines for some patients
- Buspirone is generally reserved as an augmenting agent to treat anxiety

Suggested Reading

Apter JT , Allen LA . Buspirone: future directions . *J Clin Psychopharmacol* 1999 ;**19** :86 –93 .

.....

Mahmood I , Sahaiwalla C . Clinical pharmacokinetics and pharmacodynamics of buspirone, an anxiolytic drug . *Clin Pharmacokinet* 1999 ;**36** :277–87 .

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Pecknold JC . A risk-benefit assessment of buspirone in the treatment of anxiety disorders . *Drug Saf* 1997 ;**16** :118–32 .

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Sramek JJ , Hong WW , Hamid S , Nape B , Cutler NR . Meta-analysis of the safety and tolerability of two dose regimens of buspirone in patients with persistent anxiety . *Depress Anxiety* 1999 ;**9** :131–4 .

Caprylidene

Therapeutics

Brands

- Axona
- *see index for additional brand names*

Generic?

No

Class

- Medical food
- Cognitive enhancer

Commonly Prescribed for

(bold for FDA approved)

- **Dietary management of metabolic processes associated with Alzheimer disease (mild to moderate)**
- Mild cognitive impairment

How the Drug Works

- Induces hyperketonemia and provides an alternative energy substrate to glucose in the brain
- Caprylidene is processed in the gut, resulting in medium-chain fatty acids that pass to the liver and undergo obligate oxidation, ultimately being formed into ketone bodies (acetoacetate and beta-hydroxybutyric acid (BHB))
- Ketone bodies cross the blood–brain barrier and are taken up by neurons and enter the mitochondria, where they increase mitochondrial efficiency
- Ketone bodies also generate ATP and increase pools of acetyl-CoA and acetylcholine

How Long Until It Works

- May begin working immediately

If It Works

- May improve or stabilize memory and cognitive function, but does not reverse the degenerative process

If It Doesn't Work

- Consider a cholinesterase inhibitor or memantine



Best Augmenting Combos for Partial Response or Treatment Resistance

- Cholinesterase inhibitors or memantine

Tests

- None for healthy individuals
- Triglyceride levels should be monitored periodically for individuals who meet multiple criteria for metabolic syndrome (i.e., elevated waist circumference, elevated triglycerides, high blood pressure, reduced fasting HDL, and/or elevated fasting glucose)

Side Effects

How Drug Causes Side Effects

- Caprylidene is processed in the gut, which may contribute to gastrointestinal side effects

Notable Side Effects

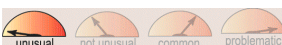
- Diarrhea, flatulence, dyspepsia
- Nausea, headache



Life-Threatening or Dangerous Side Effects

- None reported

Weight Gain



- Reported but not expected
- Some patients may experience weight loss

Sedation



- Reported but not expected

What to Do About Side Effects

- Take it with food
- Sip it slowly over approximately 30 minutes
- Consider lowering the dose

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent
- Over-the-counter simethicone, antacids, or antidiarrheals

Dosing and Use

Usual Dosage Range

- 40 g/day

Dosage Forms

- Powder 40 g/packet

How to Dose

- One 40 g packet of caprylidene should be fully mixed with 4–8 oz of liquid; dosing should occur shortly after a meal (preferably breakfast or lunch)



Dosing Tips

- Each 40 g packet of caprylidene powder contains 20 g of medium-chain triglycerides (MCTs)
- Gastrointestinal side effects are reduced if caprylidene is taken with food and/or sipped slowly over approximately 30 minutes
- Some patients may require a lower starting dose in order to improve tolerability

Overdose

- No fatalities have been reported; diarrhea, sometimes severe

Long-Term Use

- Not studied
- Drug may lose effectiveness as the course of Alzheimer disease progresses

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Absorbed in the gut; metabolized in the liver
- Crosses the blood–brain barrier



Drug Interactions

- None reported



Other Warnings/Precautions

- Use with caution in patients with known hypersensitivity to palm or coconut oil
- Use with caution in patients at risk for ketoacidosis (e.g., alcoholics, patients with poorly controlled diabetes)
- Use with caution in patients with a history of gastrointestinal inflammatory conditions

Do Not Use

- If there is a proven allergy to caprylidene, milk, or soy

Special Populations

Renal Impairment

- Not studied

Hepatic Impairment

- Not studied

Cardiac Impairment

- Not studied

Elderly

- Dose adjustment not necessary



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Controlled studies have not been conducted in pregnant women
- Animal studies have not shown teratogenic effects

Breast Feeding

- Unknown if caprylidene is secreted in human breast milk

The Art of Psychopharmacology

Potential Advantages

- Patients with residual memory problems on Alzheimer treatment

Potential Disadvantages

- Patients with gastrointestinal disorders

Primary Target Symptoms

- Memory loss in Alzheimer disease
- Memory loss in mild cognitive impairment

Pearls

- Dramatic reversal of symptoms of Alzheimer disease is not generally seen with caprylidene
- * Perhaps only 50% of Alzheimer patients are diagnosed, and only 50% of those diagnosed are treated, and then only for 200 days in a disease that lasts 7–10 years
- Must evaluate lack of efficacy and loss of efficacy over months, not weeks
- Patients who complain themselves of memory problems may have depression, whereas patients whose spouses or children complain of the patient's memory problems may have Alzheimer disease
- Treat the patient but ask the caregiver about efficacy
- What you see may depend upon how early you treat

- The first symptoms of Alzheimer disease are generally mood changes; thus, Alzheimer disease may initially be diagnosed as depression
- Women may experience cognitive symptoms in perimenopause as a result of hormonal changes that are not a sign of dementia or Alzheimer disease
- If treatment with antidepressants fails to improve apathy and depressed mood in the elderly, it is possible that this represents early Alzheimer disease
- Delay in progression in Alzheimer disease is not evidence of disease-modifying actions of caprylidene
- The most prominent side effects of caprylidene are gastrointestinal effects, which are usually mild and transient
- Weight loss can be a problem in Alzheimer patients with debilitation and muscle wasting

Suggested Reading

Henderson ST . Ketone bodies as therapeutic for Alzheimer's disease .
Neurotherapeutics 2008 ;**5** (3):470–80 .

Trull KA , Driedger A , Ingle DL , Nakhasi D. Review of the toxicologic
 properties of medium-chain triglycerides . *Food Chem Toxicol* 2000 ;**38** :79
 –98 .

Carbamazepine

Therapeutics

Brands

- Tegretol
- Carbatrol
- Equetro
- *see index for additional brand names*

Generic?

Yes (not for extended-release formulation)

Class

- Neuroscience-based Nomenclature: glutamate, voltage-gated sodium and calcium channel blocker (Glu-CB)
- Anticonvulsant, antineuralgic for chronic pain, voltage-sensitive sodium channel antagonist

Commonly Prescribed for

(bold for FDA approved)

- **Partial seizures with complex symptomatology**
- **Generalized tonic–clonic seizures (grand mal)**

- **Mixed seizure patterns**
- **Pain associated with true trigeminal neuralgia**
- **Acute mania/mixed mania (Equetro)**
- Glossopharyngeal neuralgia
- Bipolar depression
- Bipolar maintenance
- Psychosis, schizophrenia (adjunctive)

How the Drug Works

- * Acts as a use-dependent blocker of voltage-sensitive sodium channels
- * Interacts with the open channel conformation of voltage-sensitive sodium channels
- * Interacts at a specific site of the alpha pore-forming subunit of voltage-sensitive sodium channels
- Inhibits release of glutamate

How Long Until It Works

- For acute mania, effects should occur within a few weeks
- May take several weeks to months to optimize an effect on mood stabilization
- Should reduce seizures by 2 weeks

If It Works

- The goal of treatment is complete remission of symptoms (e.g., seizures, mania, pain)
- Continue treatment until all symptoms are gone or until improvement is stable and then continue treating indefinitely as long as improvement persists
- Continue treatment indefinitely to avoid recurrence of mania and seizures
- Treatment of chronic neuropathic pain most often reduces but does not eliminate pain and is not a cure since symptoms usually recur after medicine stopped

If It Doesn't Work (for Bipolar Disorder)

- * Many patients have only a partial response where some symptoms are improved but others persist or continue to wax and wane without stabilization of mood
- Other patients may be nonresponders, sometimes called treatment-resistant or treatment-refractory
- Consider increasing dose, switching to another agent, or adding an appropriate augmenting agent
- Consider adding psychotherapy
- Consider biofeedback or hypnosis for pain
- For bipolar disorder, consider the presence of noncompliance and counsel patient

- Switch to another mood stabilizer with fewer side effects or to extended-release carbamazepine
- Consider evaluation for another diagnosis or for a comorbid condition (e.g., medical illness, substance abuse, etc.)

Best Augmenting Combos for Partial Response or Treatment Resistance

- Lithium
- Atypical antipsychotics (especially risperidone, olanzapine, quetiapine, ziprasidone, and aripiprazole)
- Valproate (carbamazepine can decrease valproate levels)
- Lamotrigine (carbamazepine can decrease lamotrigine levels)
- * Antidepressants (with caution because antidepressants can destabilize mood in some patients, including induction of rapid cycling or suicidal ideation; in particular consider bupropion; also SSRIs, SNRIs, others; generally avoid TCAs, MAOIs)

Tests

- * Before starting: blood count, liver, kidney, and thyroid function tests
- During treatment: blood count every 2–4 weeks for 2 months, then every 3–6 months throughout treatment
- During treatment: liver, kidney, and thyroid function tests every 6–12 months

- Consider monitoring sodium levels because of possibility of hyponatremia
- * Before starting: individuals with ancestry across broad areas of Asia should consider screening for the presence of the HLA-B* 1502 allele; those with HLA-B*1502 should not be treated with carbamazepine

Side Effects

How Drug Causes Side Effects

- CNS side effects theoretically due to excessive actions at voltage-sensitive sodium channels
- Major metabolite (carbamazepine-10, 11 epoxide) may be the cause of many side effects
- Mild anticholinergic effects may contribute to sedation, blurred vision

Notable Side Effects

- * Sedation, dizziness, confusion, unsteadiness, headache
- * Nausea, vomiting, diarrhea
- Blurred vision
- * Benign leukopenia (transient; in up to 10%)
- * Rash



Life-Threatening or Dangerous Side Effects

- * Rare aplastic anemia, agranulocytosis (unusual bleeding or bruising, mouth sores, infections, fever, sore throat)
- * Rare severe dermatologic reactions (purpura, Stevens-Johnson syndrome)
- Rare anaphylaxis and angioedema
- Rare cardiac problems
- Rare induction of psychosis or mania
- * SIADH (syndrome of inappropriate antidiuretic hormone secretion) with hyponatremia
- Increased frequency of generalized convulsions (in patients with atypical absence seizures)
- Rare activation of suicidal ideation and behavior (suicidality)

Weight Gain



- Occurs in significant minority

Sedation



- Frequent and can be significant in amount
- Some patients may not tolerate it
- Dose-related

- Can wear off with time, but commonly does not wear off at high doses
- CNS side effects significantly lower with controlled-release formulation (e.g., Equetro, Carbatrol)

What to Do About Side Effects

- Wait
- Wait
- Wait
- Take with food or split dose to avoid gastrointestinal effects
- Extended-release carbamazepine can be sprinkled on soft food
- Take at night to reduce daytime sedation
- Switch to another agent or to extended-release carbamazepine

Best Augmenting Agents for Side Effects

- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 400–1200 mg/day
- Under age 6: 10–20 mg/kg per day

Dosage Forms

- Tablet 100 mg, 200 mg
- Tablet 100 mg chewable, 200 mg chewable
- Extended-release tablet 100 mg, 200 mg, 400 mg
- Extended-release capsule 100 mg, 200 mg, 300 mg
- Oral suspension 100 mg/5mL (450 mL)

How to Dose

- For bipolar disorder and seizures (ages 13 and older): initial 200 mg twice daily (tablet) or 1 teaspoon (100 mg) 4 times a day (suspension); each week increase by up to 200 mg/day in divided doses (2 doses for extended-release formulation, 3–4 doses for other tablets); maximum dose generally 1200 mg/day for adults and 1000 mg/day for children under age 15; maintenance dose generally 800–1200 mg/day for adults; some patients may require up to 1600 mg/day
- Seizures (under age 13): see Children and Adolescents
- Trigeminal neuralgia: initial 100 mg twice daily (tablet) or 0.5 teaspoon (50 mg) 4 times a day; each week increase by up to 200 mg/day in divided doses (100 mg every 12 hours for tablet formulations, 50 mg 4 times a day for suspension formulation); maximum dose generally 1200 mg/day
- Lower initial dose and slower titration should be used for carbamazepine suspension

Dosing Tips

- Higher peak levels occur with the suspension formulation than with the same dose of the tablet formulation, so suspension should generally be started at a lower dose and titrated slowly
- Take carbamazepine with food to avoid gastrointestinal effects
- * Slow dose titration may delay onset of therapeutic action but enhance tolerability to sedating side effects
- Controlled-release formulations (e.g., Equetro, Carbatrol) can significantly reduce sedation and other CNS side effects
- Should titrate slowly in the presence of other sedating agents, such as other anticonvulsants, in order to best tolerate additive sedative side effects
- * Can sometimes minimize the impact of carbamazepine upon the bone marrow by dosing slowly and monitoring closely when initiating treatment; initial trend to leukopenia/neutropenia may reverse with continued conservative dosing over time and allow subsequent dosage increases with careful monitoring
- * Carbamazepine often requires a dosage adjustment upward with time, as the drug induces its own metabolism, thus lowering its own plasma levels over the first several weeks to months of treatment
- Do not break or chew carbamazepine extended-release tablets as this will alter controlled-release properties

Overdose

- Can be fatal (lowest known fatal dose in adults is 3.2 g, in adolescents is 4 g, and in children is 1.6 g); nausea, vomiting, involuntary movements, irregular heartbeat, urinary retention, trouble breathing, sedation, coma

Long-Term Use

- May lower sex drive
- Monitoring of liver, kidney, thyroid functions, blood counts, and sodium may be required

Habit Forming

- No

How to Stop

- Taper; may need to adjust dosage of concurrent medications as carbamazepine is being discontinued
- * Rapid discontinuation may increase the risk of relapse in bipolar disorder
- Epilepsy patients may seize upon withdrawal, especially if withdrawal is abrupt
- Discontinuation symptoms uncommon

Pharmacokinetics

- Metabolized in the liver, primarily by CYP450 3A4